

Project 1 — Team Proposal (Refreshed, Milestone 1)

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Course: EI 7615 — Deep Learning Architectures

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Project

Discriminative computer-vision pipeline for celebrity identification and detection (Project 1). Milestone 1 establishes a single-face classification baseline over a small CelebA identity subset; Milestones 2-3 extend it to multi-face detection with YOLOv8.

Celebrity identities (final list)

Selected 4 identities — the CelebA IDs each team member individually claimed for the class-wide identity-pooling task (23-25 images each), rather than an independently-diversity-selected set:

Identity ID	# Images	Claimed by	Dominant attributes
3	25	Abdellah Faleh	Male, Wearing_Hat
7	24	Jin-Woo Hong	Male, Black_Hair
1212	25	Samuel Tong	Male, Black_Hair
8335	25	Tristan Lyons	Male, Brown_Hair

Framework choice

PyTorch (torchvision for the pretrained ResNet18 backbone). Chosen for: flexible custom nn.Module definitions for the from-scratch CNN; torchvision.models gives ImageNet-pretrained weights for the transfer-learning baseline with feature-extraction and fine-tuning modes built in; runs on both CPU (Northeastern Explorer) and any team member's local GPU without code changes.

Division of labor

Task	Owner
Data pipeline, model training (Custom CNN + ResNet18)	Abdellah Faleh
Results write-up (proposal, training-results docs)	Abdellah Faleh
GitHub repo setup & maintenance, reproducibility (README, environment)	Jin-Woo Hong
Evaluation & review (results, confusion matrix/metrics sanity-check, proposal review)	Samuel Tong
Milestone 2 planning (synthetic multi-celebrity dataset, YOLO annotations), Canvas submission coordination	Tristan Lyons

Anticipated risks

- **Compute constraints:** Explorer's CPU-only nodes (some without AVX2) can silently kill TensorFlow/PyTorch kernels during model-building ops; mitigated by keeping image size (128x128) and batch size small, and by testing on a different partition/node if a kernel dies with no traceback.
- **Class imbalance:** CelebA identities have uneven image counts; mitigated by selecting identities within a bounded image-count range and stratifying the train/val/test split per identity.
- **Small per-identity sample size:** only 30 images per identity limits how much a from-scratch CNN can learn. This risk materialized during training: an initial custom CNN configuration (heavier augmentation, higher learning rate) collapsed to predicting a single class (20% accuracy, equivalent to random guessing). We mitigated it by lightening augmentation, lowering the learning rate, and reducing dropout, which brought the custom CNN to a genuine (if lower, on this harder

identity subset) 50% test accuracy — below the fine-tuned ResNet18's 87.5%, but well above the 25% random-guessing baseline. See docs/training_results.md for the full comparison.

- **Milestone 2 dependency:** the model carried forward from Milestone 1 becomes part of the Milestone 2/3 detection pipeline, so architecture/checkpoint format decisions made here should stay compatible with a YOLOv8 integration later.